ARIJIT SIKDER

mearijitsikder@gmail.com +91 7029791801 Rajballav Pur, North 24 PGS, WB-743289, INDIA
ArijitsPy



EDUCATION

Master of Science in Physics

Ramakrishna Mission Vivekananda Educational and Research Institute

August 2022 - May 2024 Belur Math, Howrah, WB- 711202

- CGPA : 9.27, Percentage : 92.7 %
- Core Papers : Classical Mechanics, Quantum Mechanics I & II, Classical Theory of Fields I: Electrodynamics, Mathematical Tools in Theoretical Physics I & II, Computational Physics, Classical Theory of Fields II: General Relativity, Statistical Mechanics I, General Physics Lab, Atomic & Molecular Physics, Quantum Field Theory I & II, Design & Fabrication Lab, Condensed Matter Physics I & II, Particle Physics I.
- Audit : Artificial Intelligence and Physics

Bachelor of Science (Honors) in Physics

Barasat Govt. College

August 2019 - June 2022 Barasat, Kol- 700124, WB

- Core Papers : Physics, General electives: Mathematics, Chemistry under West Bengal State University.
- CGPA : 9.66, Percentage : 84.9 %

Higher Secondary

Rajballavpur High School (H.S)

2019 Rajballav Pur, 24 PGS (N), WB- 743289

- Subjects : Mathematics, Physics, Chemistry, Bio-logical Sciences, Bengali and English under WBCHSE board
- Percentage : 88.2 %

Secondary

Rajballavpur High School (H.S)

2017 Rajballav Pur, 24 PGS (N), WB- 743289

- Passed Madhyamic under WBBSE board
- Percentage : 93.86 %

ABOUT

I am a Research Scholar in the department of Physics at IIT Kharagpur, India. I am working under the supervision of Prof. Sonjoy Majumder with reasearch topic Stochastic Bose Einstein Condensation.

LANGUAGES

Bengali
English
Hindi

SKILLS

Programming Languages

Python
Fortran
C++
HTML
LaTeX

Software & Tools

Arduino IDE
OriginLab
Mathematica
Matlab
OrCAD PSpice

Operating Systems

Windows
Linux

ACHIEVEMENTS

JEST

- Qualified JEST 2025 with AIR-28 having 99.27 percentile.

GATE

- Qualified GATE 2025 with AIR-78 and GATE Score 683, Scored 49.0 out of 100.

JOINT CSIR UGC NET

- Qualified Assistant Professor(LS) June 2025 with AIR-89 having percentile 97.084.

EXPERIENCE

M.Sc. Project

A Review of the Analytical Properties of the S-Matrix

August 2023 - May 2024 RKMVERI, Belur

- The S-Matrix, or Scattering matrix, plays a crucial role in quantum field theory and particle physics by describing how initial states of a physical system evolve into final states after a scattering process. This review examines the analytical properties of the S-Matrix, focusing on its pole structure, analyticity, and unitarity. Our findings highlight the importance of the S-Matrix in understanding particle interaction and provide insights into ongoing research in theoretical physics. This work aims to bridge the gap between abstract mathematical concepts and their practical applications in particle physics.
- For details: link

Fabrication Laboratory Project

POV LED Pendulum Clock Using Arduino Nano

December 2023 RKMVERI, Belur

- In our FabLab project, we built a prototype of a Persistence of Vision (POV) LED display that features both digital and analog clock views, along with an editing feature controlled by an IR sensing remote.

Workshop

Ergodicity and its breaking

16-18 Mar 2023 RKMVERI, Belur, Howrah

Topics:
•Eigenstate Thermalisation Hypothesis (ETH), •Violations of ETH in non-integrable quantum system & gauge theories, •Thermalisation in soluble quantum theories & conformal field theories, •Thermalisation & Ergodicity in holography.

SCHOLARSHIPS

Swami Vivekananda Merit-cum-Means Scholarship

M.Sc.

2022 - 2024 RKMVERI

Swami Vivekananda Merit-cum-Means Scholarship

B.Sc.

2019 - 2022 BGC

WORKING EXPERIENCE

Physics Teacher

Vivekdisha

2024-ongoing RKMVERI, Belur, Howrah

Vivekdisha is a unique Online class program for rural school students of different schools of West Bengal conducted by RKMVERI supported by Ernst & Young Foundation (India).
Work as a Physics Teacher for Class 11 & 12.

Freelance Subject Matter Expert

Advance Physics

2022 Chegg India Private Limited

Have solved University level questions uploaded on Chegg as a Q & A Subject Matter Expert

GITHUB PROJECTS

Exam Interface For Computer Based MCQ/MSQ Tests

This project provides a basic offline exam interface for users to take exams on their local computers.
Key Features :
•Offline Functionality: The exam can be taken entirely offline in a web browser. •User-Friendly Interface: Intuitive interface with question navigation, timer, and answer marking features. •PDF Support: Allows users to upload and view PDFs related to the exam. •CSV Export: Generates a CSV file of exam responses for user review and analysis. •Data Privacy: Exam data (user responses, timings) remains local to the user's computer.
Link: link